
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a cornerstone of the safe and reliable de-
2 ployment of machine learning systems in open-world environments, where models
3 are routinely confronted with inputs drawn from classes wholly absent at training
4 time. A promising line of work sharpens OOD detection by leveraging unlabeled
5 in-the-wild data; such data, however, is intrinsically difficult to exploit, since it
6 arrives as an unlabeled mixture of in-distribution (InD) and OOD samples in which
7 the absence of any cleanly delineated set of OOD examples obstructs the training
8 of an effective OOD classifier. We introduce Medix, a framework that resolves this
9 difficulty by identifying candidate outliers directly from unlabeled data through
10 the median operation. Our reliance on the median is deliberate and principled: as
11 a robust estimator of central tendency, it is provably insensitive to the noise and
12 OOD contamination that pervade wild data. The outliers it isolates, in conjunction
13 with labeled InD data, then supply the training signal for a robust OOD classifier.
14 Theoretically, we derive two-sided error bounds establishing that Medix attains a
15 low misclassification rate for both inliers and outliers under mild assumptions; our
16 experiments decisively corroborate these guarantees, as Medix consistently and
17 substantially outperforms existing methods across a broad range of open-world
18 settings, confirming the validity of our theoretical analysis.

19 1 Introduction

20 The deployment of machine learning models in real-world applications inevitably exposes them to
21 out-of-distribution (OOD) inputs, posing acute challenges for safety and reliability. Such inputs,
22 originating from categories the model was never trained to recognize, must not be assigned confident
23 predictions; yet neural networks are inherently ill-equipped for this task, typically lacking any mecha-
24 nism by which to detect OOD inputs and appropriately abstain at deployment time (Nguyen et al.,
25 2015). Reliable OOD detection is therefore a prerequisite for trustworthy open-world deployment.

26 Identifying OOD samples at inference is at once critical and intrinsically difficult: because models
27 are never exposed to unknown distributions during training, they cannot, unaided, reliably separate
28 OOD from in-distribution (InD) data. To confront this challenge, a recent line of work (Katz-Samuels
29 et al., 2022a; Du et al., 2024a) has turned to auxiliary “in-the-wild” data. In particular, Katz-Samuels
30 et al. (2022a) regularize model training with unlabeled wild data while retaining the objective of
31 classifying labeled InD data. The appeal of wild data is twofold: it is readily available—collected
32 essentially for free once a model is operating in its target environment—and it lets the model absorb
33 the true distribution of OOD inputs encountered at test time, yielding substantially more robust OOD
34 detection.

35 Exploiting unlabeled wild data, however, is far from straightforward, precisely because such data
36 arrives as an entangled mixture of InD and OOD samples. The absence of any clean, delineated
37 set of OOD examples fundamentally complicates the design of robust detectors: the OOD detector

38 observes only samples drawn from this mixed distribution and is never informed whether a given
 39 sample is InD or OOD. This regime remains substantially underexplored, leaving considerable room
 40 for methodological advancement. Compounding the gap, few studies furnish a formal theoretical
 41 foundation; to the best of our knowledge, Du et al. (2024a) stands as the sole prior work to provide
 42 such a foundation for the “in-the-wild” setting, underscoring the need for principled treatments of
 43 this problem.

44 In parallel, median-based estimation has recently proven highly effective in data pruning (Acharya
 45 et al., 2024), owing to its intrinsic robustness against corrupted or atypical samples. These two
 46 threads—the open problem of exploiting unlabeled wild data and the demonstrated robustness of
 47 median-based estimation—converge naturally, and motivate the central question this paper sets out to
 48 answer:

How can median-based methods leverage unlabeled wild data to facilitate OOD
 detection with theoretical guarantees?

49 We answer this question decisively in the affirmative by introducing a median-centric perspective on
 50 OOD detection. Concretely, we formulate a median-based optimization framework and develop an
 51 accompanying algorithm, Medix, that identifies OOD samples within wild data at a demonstrably
 52 low error rate. We establish a theoretical foundation that guarantees this low error and validate it
 53 empirically, demonstrating both the robustness and the efficiency of the algorithm. In doing so,
 54 we furnish one of the few rigorous theoretical foundations for the unlabeled “in-the-wild” setting.
 55 Specifically, we prove that median-based filtering is provably robust for outlier detection in unlabeled
 56 mixtures by bounding the fraction of InD samples erroneously flagged as outliers. This fraction is
 57 governed by two complementary and interpretable effects: a *contamination effect*, which quantifies
 58 the influence of OOD points and remains controlled whenever the OOD proportion stays below 50%;
 59 and a *concentration effect*, which exploits the sub-Gaussian nature of InD gradients to bound the
 60 deviation of individual InD points from their mean gradient. Acting jointly, these two effects guarantee
 61 that, with high probability, the number of misclassified InD points remains small—establishing the
 62 reliability of the method even under substantial, near-worst-case contamination.

63 We benchmark Medix against two families of methods: (1) those trained solely on InD data, and
 64 (2) those trained on both InD data and an auxiliary unlabeled dataset. On CIFAR-100 (Krizhevsky
 65 et al., 2009), Medix improves markedly over the strong baseline KNN+ (Sun et al., 2022), lowering
 66 average FPR95 by 40.98%. Crucially, whereas approaches such as Outlier Exposure (Hendrycks
 67 et al., 2019) depend on a clean auxiliary dataset—imposing the stringent distributional assumption
 68 that the auxiliary data is fully separable from the InD data—Medix attains superior results without
 69 invoking any such assumption, and is therefore applicable to a substantially broader range of realistic
 70 deployment settings. Relative to the state-of-the-art wild-data method WOODS (Katz-Samuels et al.,
 71 2022a), Medix further reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10,
 72 establishing a consistent advantage over the strongest competing approach in both regimes.

73 We summarize our key contributions as follows:

- 74 **C1) A median-centric filtering framework.** We propose Medix, a median-centric greedy
 75 method that first filters outliers from unlabeled wild data and then trains an OOD detector
 76 on the extracted outliers together with the labeled InD samples. The gradient-based median
 77 filtering stage constitutes our central methodological contribution and is what enables robust
 78 OOD detection without a clean auxiliary set.
- 79 **C2) Two-sided theoretical guarantees.** We establish provable, two-sided guarantees for the ro-
 80 bustness of median-based filtering in jointly identifying inliers and outliers within unlabeled
 81 mixtures. Concretely, we prove that the misclassification rates for both InD samples flagged
 82 as outliers and OOD samples retained as inliers are tightly controlled by two interpretable
 83 effects: a contamination effect, which remains bounded whenever the OOD proportion
 84 is below 50%, and a concentration effect, which follows from the stable, sub-Gaussian
 85 gradient behavior of InD samples. To our knowledge, this constitutes one of the few rigorous
 86 foundations available for the unlabeled in-the-wild setting.
- 87 **C3) Comprehensive empirical validation.** We conduct an extensive evaluation of Medix
 88 across eleven InD–OOD pairs and against 20 competitive baselines, and Medix outperforms
 89 every one of them, delivering uniformly superior detection across the board. We further

90 demonstrate that Medix outlier extraction attains a strikingly low error rate (e.g., only 12.5%;
 91 see Figure 2), directly and quantitatively corroborating our theoretical findings.

92 2 Preliminaries and Problem Setup

93 In this section we formalize the OOD detection problem and then precisely specify the data setup,
 94 model architecture, loss functions, and learning objective that underpin the remainder of the paper.

95 **Labeled In-Distribution Data.** Let $\mathcal{X}$ denote the input space and $\mathcal{Y} = \{1, \dots, K\}$ the label
 96 space; jointly, these two spaces characterize the structure of the InD data. A labeled dataset $\mathcal{S}_{\text{in}} =$
 97 $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ comprises n pairs drawn independently and identically distributed (i.i.d.)
 98 from $\mathbb{P}_{XY}$, an unknown joint distribution over $\mathcal{X} \times \mathcal{Y}$. We write $\mathbb{P}_{\text{in}}$ for the marginal of $\mathbb{P}_{XY}$ on
 99 $\mathcal{X}$ —that is, the underlying distribution of InD inputs—and we train the InD model on $\mathcal{S}_{\text{in}}$.

100 **Out-of-distribution detection.** We consider the practical scenario in which a model trained on
 101 labeled InD data is subsequently deployed in environments that may present OOD inputs from classes
 102 unseen during training, i.e., inputs carrying some label $y \notin \mathcal{Y}$. On such inputs the model must abstain
 103 rather than commit to a class prediction. The central objective at inference time is therefore to decide
 104 whether a given input is drawn from the InD distribution or instead originates from an OOD source.

105 **Unlabeled wild data.** A principal obstacle in OOD detection is the scarcity of labeled OOD samples:
 106 the OOD sample space is potentially unbounded, rendering the collection of labeled examples both
 107 costly and impractical. To circumvent this obstacle, and following Katz-Samuels et al. (2022a),
 108 we incorporate unlabeled wild data, $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$, into our learning framework, thereby
 109 faithfully reflecting real-world deployment. Wild data constitutes a blend of InD and OOD samples
 110 and is readily gathered once a model pre-trained on $\mathcal{S}_{\text{in}}$ is put into operation. Consistent with Du et al.
 111 (2024a); Katz-Samuels et al. (2022a), we adopt the Huber contamination model (Huber, 1964) to
 112 characterize the marginal distribution of the wild data,

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

113 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ is the OOD distribution over $\mathcal{X}$. The boundary
 114 case $\pi = 0$ corresponds to the complete absence of OOD samples and renders the problem trivial; we
 115 therefore restrict attention to the informative regime $\pi \in (0, 1]$.

116 **Models and Loss Functions.** Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ denote the InD classifier parameterized by $\phi \in \Phi$,
 117 with Φ its parameter space; the output of f_ϕ is a soft probability distribution over the $K = |\mathcal{Y}|$ InD
 118 classes, and its supervised loss is $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection we introduce a separate
 119 classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$, parameterized by $\theta \in \Theta$ over parameter space Θ , whose associated binary
 120 loss is $\ell_b(g_\theta(x), y_b)$ with $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$. Here the positive value $y_+ > 0$ encodes the InD
 121 class and the negative value $y_- < 0$ the OOD class, so that the sign of g_θ directly yields the detection
 122 decision at inference.

123 **Learning objective.** Our framework jointly trains the OOD detector g_θ and the multi-class classifier
 124 f_ϕ , drawing simultaneously on the InD data $\mathbb{P}_{\text{in}}$ and the wild data $\mathbb{P}_{\text{wild}}$. At test time we assess
 125 performance with the following complementary metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{XY}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

126 where $\mathbb{I}\{\cdot\}$ is the indicator function and $\mathbb{P}_{\text{out}}^{\text{test}}$ is the OOD test distribution. FPR and TPR jointly
 127 characterize the quality of OOD detection, while $\text{Acc}(f_\phi)$ certifies that InD classification accuracy is
 128 preserved rather than sacrificed for detection performance.

129 3 Method: Median-Centric Framework for OOD Detection

130 We now present **Medix**, a learning paradigm that performs OOD detection by systematically har-
 131 nassing unlabeled wild data. By construction, Medix transcends the limitations of conventional
 132 approaches confined to InD data, and it is particularly well-suited to open-world deployment, where
 133 models are routinely confronted with previously unseen inputs. The framework decomposes into

two tightly coupled stages: **1) Outlier Extraction**, a filtering procedure that isolates candidate OOD samples from the unlabeled wild data (Section 3.1); and **2) Detector Training**, in which a binary OOD detector is trained on the InD data together with the outlier candidates identified in stage 1 (Section 3.2), following the protocol of Du et al. (2024a). As the following sections establish, this two-stage design not only extracts OOD data from the unlabeled wild set effectively, but also furnishes a robust and theoretically grounded foundation for deploying machine learning models in dynamic, open-world environments.

3.1 Extracting Candidate Outliers from the Wild Data

To isolate candidate outliers from the wild mixture $\mathcal{S}_{\text{wild}}$, Medix adopts an optimization-based formulation grounded in the parameter gradients of a classification model f_ϕ trained exclusively on the InD dataset $\mathcal{S}_{\text{in}}$. The underlying rationale is precise: the gradient of the InD loss evaluated at a sample, taken with respect to the parameters of an InD-trained model, quantifies how far that sample departs from InD behavior, and thereby supplies a discriminative signal for filtering. We formalize this procedure below.

Reference gradient estimation from InD data. The first step is to establish a reference gradient from the InD dataset $\mathcal{S}_{\text{in}}$. We train the classifier f_ϕ on $\mathcal{S}_{\text{in}}$ via empirical risk minimization (ERM),

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \text{ where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. With the classifier trained, we define the reference gradient $\bar{\nabla}_{\text{in}}$ as the empirical mean, over the InD data, of the loss gradients with respect to the model parameters:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

The reference gradient $\bar{\nabla}_{\text{in}}$ thus functions as a stable anchor in gradient space against which the deviation of any other data point can be quantified.

Motivation. The design of Medix rests on a precise and directly testable hypothesis: as the proportion of OOD samples in the wild dataset $\mathcal{S}_{\text{wild}}$ grows, the aggregate gradient statistic of $\mathcal{S}_{\text{wild}}$ should deviate increasingly from the reference InD gradient $\bar{\nabla}_{\text{in}}$. To probe this hypothesis directly, we conduct a controlled experiment with CIFAR-10 (Krizhevsky et al., 2009) as the InD dataset and SVHN (Netzer et al., 2011) as the OOD dataset. We initialize $\mathcal{S}_{\text{wild}}$ with 10,000 CIFAR-10 samples held strictly disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to compute $\bar{\nabla}_{\text{in}}$), then incrementally inject SVHN OOD samples while tracking the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the element-wise median (EWM) of the wild-set gradients:

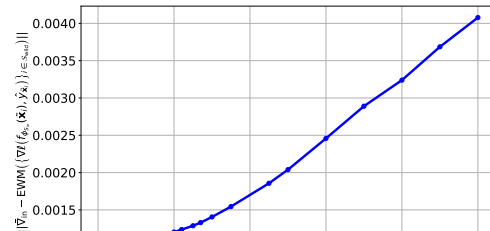
$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}_{\text{wild}}} \right) \right\|.$$

As shown in Figure 1, the L_2 -norm deviation increases clearly and monotonically with the OOD count, unambiguously confirming our hypothesis. This monotonic relationship is far more than a diagnostic observation: it directly motivates the algorithm developed in the next section and, in particular, its stopping criterion, whereby the procedure terminates once the inter-iteration change in L_2 -norm deviation falls below a threshold ϵ —a mechanism we detail below.

Filtering potential outliers from unlabeled wild data. Building on the monotonic behavior established in Figure 1, we cast outlier identification as the following optimization problem, whose solution yields the outlier subset $\mathcal{S}_{\text{out}}^*$ in $\mathcal{S}_{\text{wild}}$:

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

Here $\text{EWM}(\cdot)$ denotes the element-wise median function and $\hat{y}_{\tilde{\mathbf{x}}_i}$ the predicted label of a wild sample $\tilde{\mathbf{x}}_i$; for notational brevity we write $G_{\mathcal{S}} = \{\nabla \ell(\tilde{\mathbf{x}}_i)\}_{i \in \mathcal{S}}$. Intuitively, (4) seeks the subset $\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}$ whose gradient EWM lies closest to the reference gradient



175 $\bar{\nabla}_{\text{in}}$. Given the monotonic relationship established
 176 in Figure 1, the minimizer $\mathcal{S}_{\text{in}}^*$ is expected to re-
 177 cover precisely the InD portion of $\mathcal{S}_{\text{wild}}$, so that its
 178 complement $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ isolates the OOD
 179 portion.

180 Because exhaustively solving (4) is combinatorial
 181 and hence computationally prohibitive as $|\mathcal{S}_{\text{wild}}|$
 182 grows, we introduce a tractable greedy approxima-
 183 tion based on a leave-one-out criterion, formalized
 184 in Algorithm 1. The algorithm performs iterative
 185 outlier detection by quantifying, at each step, how
 186 the removal of an individual wild sample reshapes
 187 the gradient EWM relative to the InD reference.

188 At each iteration the algorithm computes the EWM
 189 of the current wild-set gradients and compares
 190 it against the reference gradient $\bar{\nabla}_{\text{in}}$. Let $d_t =$
 191 $\|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ denote the L_2 distance between
 192 $\bar{\nabla}_{\text{in}}$ and the EWM of the gradients remaining in the wild set $\mathcal{S}$. For each candidate $i \in \mathcal{S}$, the algorithm
 193 evaluates the leave-one-out EWM $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$ and the resulting distance $\|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$,
 194 yielding the induced reduction $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$; the k samples that induce the
 195 largest reductions—and hence contribute most to the deviation from the InD reference—are flagged as
 196 OOD and removed. This process iterates until no remaining sample yields a significant reduction δ_i or
 197 a maximum iteration count is reached. Formally, the convergence criterion requires the inter-iteration
 198 change in L_2 distance to fall below a predefined threshold ϵ . This criterion is a direct consequence
 199 of the monotonic trend observed in Figure 1: once a substantial fraction of OOD samples has been
 200 removed, the L_2 distance settles to a small, stable value, precisely marking the point at which the
 201 algorithm should halt so as to avoid erroneously excising genuine InD samples and thereby degrading
 202 downstream performance.

Algorithm 1 Iterative Outlier Detection via **Medix**

Require: $\bar{\nabla}_{\text{in}}, G_i = \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})$, maximum iterations T , hyperparameters ϵ, k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$
 (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

203 3.2 Training the OOD Detector with Candidate Outliers

204 Having extracted the candidate outlier set $\hat{\mathcal{S}}_{\text{out}}$ from the wild data, we train the OOD detector g_{θ} to
 205 maximally separate InD data from the extracted candidate outliers, following the protocol of Du et al.
 206 (2024a). The objective enforces separability at the decision boundary (thresholded at 0), driving g_{θ}
 207 toward positive outputs on labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ and negative outputs on candidate outliers
 208 $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$. Concretely, the loss is

$$\begin{aligned} \mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_{\theta}) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_{\theta}) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_{\theta}(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_{\theta}) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_{\theta}(\tilde{\mathbf{x}}) > 0\}. \end{aligned} \tag{5}$$

Because the 0/1 loss is non-differentiable, we replace it with a differentiable sigmoid-based binary surrogate, which preserves alignment with the original objective while rendering the optimization tractable via gradient-based methods. The framework simultaneously retains the InD risk of Equation 2, thereby safeguarding predictive accuracy on InD samples even as detection is optimized. This unified objective substantially strengthens the generalization of g_θ and enables robust detection of OOD samples drawn from $\mathbb{P}_{\text{out}}$, as our experiments in Section 5 confirm.

4 Theoretical Analysis

We now formalize the theoretical guarantees underlying the filtering stage of Medix. The two theorems below furnish provable, non-asymptotic upper bounds on the misclassification rates for InD and OOD points, respectively, and, taken together, they establish a two-sided robustness guarantee for EWM filtering: neither are too many inliers discarded nor too many outliers retained. Complete proofs are deferred to Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 establishes that the inlier misclassification rate is governed by exactly two effects: a *concentration term*, which captures statistical fluctuations in the InD gradients and vanishes as the sample size grows, and a *contamination term*, which quantifies the influence of OOD points on the EWM. Decisively, the bound remains controlled for any contamination ratio $\pi < 0.5$ —a property that directly certifies the robustness of median-based filtering even under substantial OOD presence. This naturally raises the complementary question: *how many true outliers evade the filter and are mistakenly retained as inliers?* The following theorem settles this question by bounding the OOD misclassification rate.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1-\pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 complements the preceding result by bounding the fraction of OOD points mistakenly retained as InD. Together, Theorems 4.1 and 4.2 constitute a complete two-sided robustness guarantee for Medix filtering, in which the inlier and outlier misclassification rates are jointly governed by the balance among three interpretable effects:

Contamination Effect. In Theorem 4.1, the median-based criterion remains provably stable even when the wild dataset contains a large fraction π of OOD points, provided only that $\pi < 0.5$. This behavior is encoded by the term $\pi/[2(1-\pi)]$, which grows with π and precisely captures the risk that OOD gradients skew the EWM. Symmetrically, Theorem 4.2 carries the mirror-image penalty $\frac{1-\pi}{2\pi}$, which quantifies the reciprocal difficulty of isolating OOD points when they are the underrepresented population.

Concentration Effect. Both bounds exploit the sub-Gaussian structure of the gradient coordinates. For InD data, this structure guarantees that the bulk of gradients concentrate tightly near $\bar{\nabla}_{\text{in}}$, keeping

the median stable even at finite sample sizes; for OOD data, concentration around μ_{out} enables us to quantify the risk that a misaligned OOD sample slips past the filter. Both concentration terms decay at the parametric rate $1/\sqrt{m}$, with m the number of samples from the corresponding class, and therefore vanish in the large-sample limit.

Separation Effect. Unique to Theorem 4.2, the separation effect specifies how far the OOD mean gradient must lie from the InD mean gradient for OOD samples to be reliably rejected. The exponential factor $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ encodes this trade-off exactly: the greater the separation between the two gradient distributions, the exponentially smaller the probability that an OOD gradient deceives the filter.

Theorems 4.1 and 4.2 jointly establish that, with high probability, median filtering achieves robust separation of InD and OOD samples in unlabeled mixtures. The fraction of InD samples misclassified as outliers is bounded by the contamination and concentration effects, while the fraction of OOD points incorrectly retained is bounded by an exponential separation term, a concentration term, and a reverse contamination effect. Together, these results furnish rigorous theoretical assurance that Medix simultaneously controls both error types under mild and empirically verifiable assumptions.

Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates. The coordinate-wise sub-Gaussianity of InD gradients is the linchpin of our concentration bounds and, more broadly, a standard cornerstone of robust statistical estimation. This assumption is strongly supported by the empirical evidence. Figure 4a shows that the histogram of InD gradient values is bell-shaped and tightly concentrated about the mean, exhibiting precisely the light, exponentially decaying tails that characterize sub-Gaussian random variables and that render extreme gradient values rare. Figure 4b reinforces this conclusion via a Q-Q plot (Wilk and Gnanadesikan, 1968) comparing the empirical gradient quantiles against the theoretical Gaussian quantiles: the near-exact alignment with the 45-degree reference line indicates that the empirical distribution closely tracks a Gaussian, and since Gaussians are a subclass of sub-Gaussian variables, the assumption is directly and empirically justified. *Moreover, we establish a more general variant of these bounds that dispenses with the sub-Gaussian assumption entirely, requiring only bounded second and fourth moments; this appears as Theorem C.3 (Appendix C.3). While the rates degrade gracefully in this weaker regime, the core robustness guarantee of Medix persists intact, demonstrating that our conclusions extend well beyond the sub-Gaussian setting.*

5 Experiments

This section establishes the efficacy of Medix across a broad range of InD–OOD dataset pairs, benchmarking it comprehensively against 20 widely-used baselines. All experiments are conducted on NVIDIA A100-SXM4-80GB GPUs, and the code required to reproduce our results will be released upon acceptance.

5.1 Models, Datasets, and Baselines

Datasets. To ensure a rigorous and fair comparison with prior work, we adopt the exact experimental protocol of Katz-Samuels et al. (2022a), the study that introduced the problem of learning OOD detectors from wild data. We use CIFAR-10 and CIFAR-100 as InD datasets ($\mathbb{P}_{\text{in}}$), and for OOD testing we employ a diverse suite of natural-image datasets: PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE & LSUN-C (Yu et al., 2015) ($\mathbb{P}_{\text{out}}$). To simulate wild data ($\mathbb{P}_{\text{wild}}$), we mix a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) at a default contamination level $\pi = 0.5$ —the most demanding balanced-mixture regime admitted by our theory. Concretely, when PLACES365 serves as the OOD test set, we form the wild mixture by combining CIFAR with PLACES365 and evaluate on PLACES365; this procedure is repeated identically for every OOD dataset and every baseline, ensuring strict comparability. Each InD CIFAR set is partitioned into two disjoint halves: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, while the remainder form the wild mixture $\mathcal{S}_{\text{wild}}$. Gradients are computed on the penultimate-layer weights, which prior work has established to be particularly informative for OOD detection (Huang et al., 2021a).

Evaluation metrics. We report three standard, complementary metrics: (1) the False Positive Rate (FPR95 $\downarrow$) of OOD samples at a 95% True Positive Rate on InD samples, (2) the Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification Accuracy (InD Acc $\uparrow$). The first two quantify the quality of OOD detection, while the third verifies that InD classification performance is preserved rather than traded away.

Baselines. Our comparison spans a broad and competitive set of baselines, systematically organized by whether they are trained on InD data alone or on both InD and wild data. Among methods trained exclusively on InD data ($\mathbb{P}_{\text{in}}$), we compare Medix against a comprehensive range of established detectors: Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all built on a softmax cross-entropy-trained model. For completeness we further include the contrastive-loss methods CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). Among methods that leverage both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019) and energy-regularization learning (Liu et al., 2020b), which regularize training by promoting lower confidence or higher energy on outlier data; against WOODS (Katz-Samuels et al., 2022a), which introduced the notion of wild unlabeled data and exploits it via constrained optimization; and against the more recent CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024), which we include to ensure a thorough and up-to-date evaluation against the current state of the art.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we instantiate the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ as a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2, trained by stochastic gradient descent with momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs under cosine learning-rate decay, with batch size 128 and dropout rate 0.3. The Medix hyperparameters ϵ and k are selected from $\{5\text{e-}5, 5\text{e-}4, 5\text{e-}3, 5\text{e-}2\}$ and $\{4\text{k}, 7\text{k}, 10\text{k}, 20\text{k}\}$ respectively, chosen with regard to dataset size so as to maximize OOD detection performance. The OOD classifier g_{θ} is initialized from the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and augmented with a linear layer that performs binary classification on the penultimate-layer features. It is subsequently fine-tuned for 100 epochs at learning rate 0.001 according to Equation (5), combining the binary classification loss with the InD classification loss at a relative weight of 10 on the binary term; all remaining training parameters are held identical to those used for $\phi_{\mathcal{S}_{\text{in}}}$.

Table 1: OOD detection performance of Medix versus all baselines with CIFAR-10 as InD data, reported per OOD test set (SVHN, PLACES365, LSUN-C, LSUN-Resize, TEXTURES) and as an average, alongside InD accuracy. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. Medix attains the best FPR95 and the best AUROC on *every* OOD test set, surpassing both the InD-only detectors and the strongest wild-data baselines while retaining competitive InD accuracy.

Methods	OOD Datasets												InD ACC $\uparrow$
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES		Average		
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	
	Using $\mathbb{P}_{\text{in}}$ only												
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	91.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{out}}$												
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

5.3 Results

Our principal results on CIFAR-100 appear in Table 2, where Medix outperforms every OOD detection baseline by a decisive margin. Two patterns stand out. First, methods that exploit both

Table 2: OOD detection performance of Medix versus all baselines with CIFAR-100 as InD data, reported per OOD test set and as an average, alongside InD accuracy. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. On this more challenging 100-class benchmark, Medix again achieves the best FPR95 and the best AUROC across every OOD test set, decisively outperforming even the strongest wild-data baselines.

Methods	OOD Datasets										InD Acc $\uparrow$		
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES			Average	
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		FPR95 $\downarrow$	AUROC $\uparrow$
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

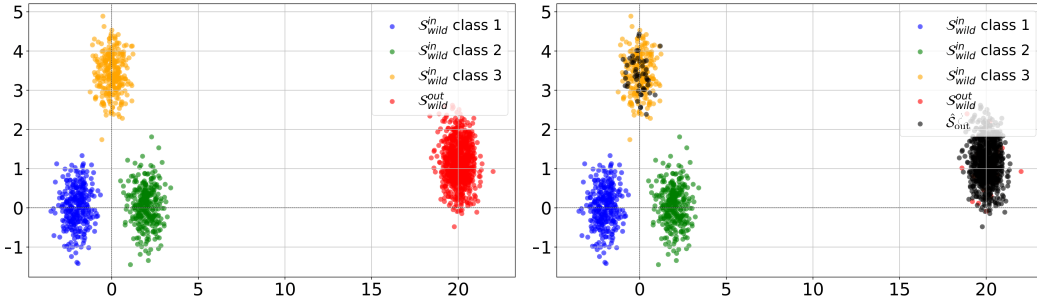


Figure 2: Medix applied to a synthetic 2D wild mixture. (a) The wild set comprises InD data $\mathcal{S}_{wild}^{in}$, drawn from three multivariate Gaussians, together with OOD data $\mathcal{S}_{wild}^{out}$. (b) The outliers $\hat{\mathcal{S}}_{out}$ flagged by Medix (black) align almost exactly with the ground-truth OOD points: only 12.5% of $\hat{\mathcal{S}}_{out}$ are in fact InD points, providing a direct visual confirmation of the low extraction error predicted by Theorems 4.1 and 4.2.

319 InD and wild data substantially surpass those confined to InD data: relative to the strong InD-only
320 baseline KNN+, Medix lowers FPR95 by 52.31% on PLACES365 and by 38.24% on TEXTURES,
321 precisely quantifying the substantial value of in-the-wild data as a regularizing signal. Second, and
322 more tellingly, Medix also dominates the strongest wild-data competitors: it attains an average
323 FPR95 of 5.42% on CIFAR-100, a 1.32% absolute improvement over the state-of-the-art WOODS,
324 while sustaining a competitive InD accuracy of 73.33%. The modest accuracy gap is fully and
325 transparently accounted for by the experimental protocol—Medix is trained on 25,000 labeled InD
326 samples, whereas the InD-only baselines use the full 50,000-sample CIFAR-100 training set—and
327 is therefore a direct consequence of the wild-data setting rather than any limitation intrinsic to the
328 method. The CIFAR-10 results in Table 1 tell an entirely consistent story: Medix surpasses all
329 baselines, improving FPR95 over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES,
330 and thereby further confirming its effectiveness at detecting OOD samples across distinct benchmarks.

331 **A representative visual example of Medix.** To render the behavior of Algorithm 1 (Outlier
332 Extraction) directly interpretable, we construct a deliberately simple two-dimensional synthetic
333 experiment. The InD data is drawn from three multivariate Gaussians—one per class—with mean
334 vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$ and shared covariance $0.25 \cdot I$, while the OOD data follows
335 $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. As Figure 2 shows, Medix recovers the outliers with near-perfect fidelity: it
336 correctly flags 87.5% of the true OOD samples, with the extracted set containing only 12.5% InD
337 contamination. This concrete instance vividly and directly substantiates the low extraction error
338 established by our theoretical analysis.

Additional studies and insights. Owing to space constraints, we defer a substantial body of further experiments to Appendix A, each of which independently reinforces the conclusions above: (1) hyperparameter-selection ablations and a sensitivity analysis (Appendix A.2) establishing the strong robustness of Medix to its hyperparameters; (2) a comparison of EWM against the geometric median (Appendix A.1), demonstrating that EWM is more sensitive to distributional shift and is therefore the more effective and reliable filtering choice; (3) a comprehensive comparison against methods trained with competitive contrastive objectives (Appendix A.3), in which Medix outperforms even these strong baselines by a significant margin; (4) an evaluation on large-scale data under the demanding unseen-OOD setting $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4), where Medix again leads by a wide margin; (5) an assessment of the computation and memory efficiency of Medix (Appendix A.6); (6) an analysis of the impact of pseudo-label quality (Appendix A.5), demonstrating resilience to noisy or low-confidence labels; and (7) a comparison with semi-supervised open-set recognition methods (Appendix A.7). Collectively, these studies confirm that the advantages of Medix are robust across models, scales, and evaluation regimes.

6 Related Work

OOD detection has attracted intense and sustained interest in recent years (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djurisic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). One prominent family derives scoring functions to gauge the distributional origin of an input, spanning distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al., 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al., 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based approaches (Bendale and Boult, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020). While effective, these post-hoc scores operate on a fixed InD-trained model and do not exploit unlabeled wild data during training, as Medix does.

A second family imposes regularization during training (Malinin and Gales, 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020; Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b, 2023; Wang et al., 2023a), for instance by suppressing model confidence (Lee et al., 2017; Hendrycks et al., 2019) or raising its energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data. Critically, however, most of these methods presuppose access to a clean auxiliary OOD dataset—a restrictive assumption that Medix explicitly dispenses with.

A related line of work relaxes the labeled-only assumption by incorporating unlabeled wild data (Zhou et al., 2021; Katz-Samuels et al., 2022b; He et al., 2023; Katz-Samuels et al., 2022a; Geng et al., 2025), yet these studies stop short of furnishing an explicit outlier-detection mechanism. Du et al. (2024a,b) do introduce explicit outlier filtering, but their thresholding technique differs fundamentally from ours: Medix instead detects outliers through a novel median-centric criterion equipped with provable guarantees. A further, structural distinction is equally important. Katz-Samuels et al. (2022a); Du et al. (2024a) assume batch-level mixing, in which each batch contains a fixed ratio of InD and OOD samples; with large outsourced datasets such structure is simply unavailable, since data is mixed randomly across the corpus. Medix removes this requirement entirely, operating directly and without modification under the more realistic regime of dataset-level mixing. Finally, many works adopt positive-unlabeled learning, training classifiers from positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al., 2022; Zhao et al., 2022; Acharya et al., 2022); the essential difference is that these methods only separate $\mathbb{P}_{\text{out}}$ from $\mathbb{P}_{\text{in}}$ and do not simultaneously train an OOD classifier. In sharp contrast, Medix couples a median-centric outlier-identification procedure—with provably low error rates—directly to the training of a dedicated OOD detector.

7 Conclusions

We introduced Medix, a median-centric framework for OOD detection that transforms unlabeled in-the-wild data into a reliable training signal. By exploiting the inherent robustness of the median operation, Medix filters outliers from mixed unlabeled data and thereby enables the training of a dependable OOD detector without any clean auxiliary set. Our theoretical analysis furnished provable bounds on the inlier misclassification rate, establishing that Medix remains robust even under substantial OOD contamination—up to 50%—with errors controlled by sub-Gaussian concentration and contamination effects, and complemented these with matching bounds on the OOD misclassification rate that guarantee accurate isolation of outliers under clear separation. Empirically, Medix surpassed all 20 baselines across diverse benchmarks: it reduced average FPR95 by 40.98% relative to strong baselines such as KNN+, decisively outperformed state-of-the-art wild-data methods including WOODS and DRL, and attained an outlier-extraction error rate as low as 12.5%. Taken together, these tightly coupled theoretical and empirical results establish median-centric filtering as a principled, robust, and highly effective foundation for OOD detection from unlabeled data.

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664 A Additional Studies

665 A.1 Element-wise Median vs Geometric Median

666 A natural question is why Medix employs the element-wise median rather than the geometric median
 667 (Acharya et al., 2024) for outlier detection. We answer it with a direct, controlled comparison.
 668 To answer it, we ran a controlled experiment with CIFAR-
 669 10 as the InD dataset and SVHN as the OOD dataset.
 670 Here $\mathcal{S}_{\text{wild}}$ consists of 5k CIFAR-10 samples held strictly
 671 disjoint from the data used to train $\phi_{\mathcal{S}_{\text{in}}}$ (and hence to
 672 compute $\bar{\nabla}_{\text{in}}$); we then incrementally add SVHN OOD
 673 samples while jointly tracking the L_2 -norm deviation be-
 674 tween $\bar{\nabla}_{\text{in}}$ and the element-wise median of the wild-set
 675 gradients and the proportion of OOD samples removed.
 676 As Figure 3 demonstrates, at any matched OOD count
 677 the element-wise median flags a substantially higher pro-
 678 portion of OOD samples as outliers than the geometric
 679 median. This establishes conclusively that the element-
 680 wise median is markedly more sensitive to distributional
 681 shift, making it the more effective and reliable filtering
 682 criterion for Medix.

683 A.2 Hyperparameter 684 Selection and Sensitivity Analysis

685 We conducted an ablation to precisely quantify the sensi-
 686 tivity of Medix to the hyperparameters ϵ and k , using
 687 CIFAR-100 as the InD dataset and SVHN as the OOD
 688 dataset. As Table 3 shows, Medix is strongly robust to
 689 variation in both ϵ and k . It attains a remarkably low FPR95 of 0.16 at $\epsilon = 0.005$, $k = 20000$,
 690 and—crucially—perturbing these settings degrades performance only gracefully rather than cata-
 691 strophically: reducing ϵ to 0.0005 or halving k to 10000 yields FPR95 values of 0.20 and 0.68
 692 respectively, each of which still comfortably surpasses every baseline. Thus, while careful tuning
 693 yields the strongest results, Medix retains its decisive advantage across a wide range of hyperpa-
 694 rameter choices. This insensitivity to exact tuning is a substantial practical asset, rendering Medix
 695 a dependable choice for real-world deployments in which exhaustive hyperparameter search is
 696 infeasible.

Table 3: Effect of the hyperparameters ϵ and k on Medix OOD detection (CIFAR-100 InD, SVHN OOD, FPR95 $\downarrow$), with representative InD-only and wild-data baselines listed for reference. Across every tested (ϵ, k) configuration, Medix remains far below all baselines, demonstrating that its advantage does not hinge on precise tuning.

Method	FPR95 $\downarrow$	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

697 A.3 Comprehensive comparison with recent competitive methods

698 We extend our comparison to a suite of competitive baselines built on diverse contrastive objec-
 699 tives—SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al.

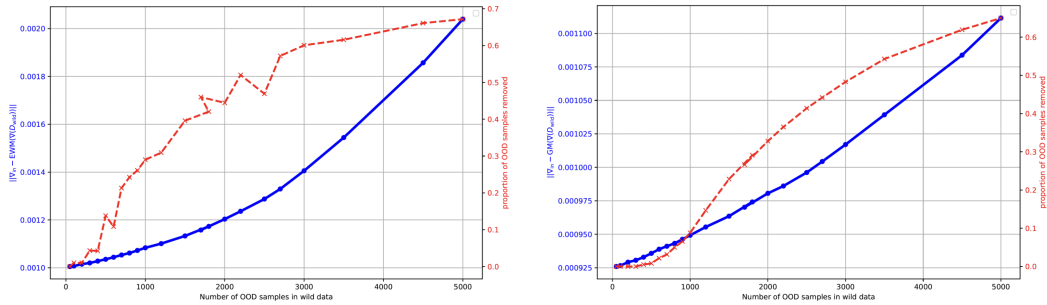


Figure 3: Element-wise median (EWM) versus geometric median (GM) as OOD samples are progressively added to $\mathcal{S}_{\text{wild}}$ (CIFAR-10 InD, SVHN OOD, Wide ResNet): at any matched OOD count, EWM flags a substantially higher proportion of true OOD samples as outliers, directly evidencing its greater sensitivity to distributional shift and motivating its adoption as the filtering criterion of Medix.

Table 4: OOD detection performance of Medix versus recent competitive baselines with CIFAR-100 as InD data (ResNet-34), reported per OOD test set and as an average. All numbers are averaged over five runs (Medix reported with standard deviations); the best result in each column is in **bold**, with lower FPR95 and higher AUROC being better. Medix leads on every single OOD test set, decisively outperforming even the strong methods trained with contrastive objectives.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.18	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

(2023)—which we include for completeness despite Medix not itself employing contrastive learning. We further add the more recent CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c) to ensure a thorough and up-to-date evaluation. For this study we use CIFAR-100 as the InD dataset and train the ResNet-34 model¹ following Ming et al. (2023); Zhang et al. (2024): stochastic gradient descent with momentum 0.9 and weight decay 10^{-4} for 500 epochs, initial learning rate 0.5 under cosine scheduling, and batch size 512, reusing the checkpoints of Ming et al. (2023)². As Table 4 shows, Medix outperforms every one of these competitive baselines by a significant margin—including those trained with contrastive objectives—achieving an average FPR95 of 8.68% and an average AUROC of 97.76%. Beyond this decisive gain in raw performance, Medix uniquely furnishes a provably low error rate, offering rigorous theoretical guarantees that the great majority of these alternatives lack.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

We now examine whether Medix scales to large-scale wild OOD data under the demanding unseen-OOD setting. This regime is challenging on two distinct counts: (1) it draws on a large-scale wild OOD dataset ($\mathbb{P}_{\text{out}}$), which sharply increases both data complexity and computational demand; and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) is deliberately rendered distributionally distinct from the training-time wild OOD data, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, thereby faithfully mirroring realistic deployment in which the OOD encountered at test time need not match what was observed during training. We take CIFAR-100 as the labeled InD data, the 300K Random Images dataset Hendrycks et al. (2019) as the unlabeled wild OOD data, and SVHN as the test OOD data. The substantial distributional gap between the large-scale wild OOD data (300K Random Images) and the OOD test data (SVHN) renders this a particularly stringent test of generalization to unseen outliers. Comparing against baselines that likewise leverage wild data during training, Table 5 shows that Medix is decisively superior, delivering both a markedly lower FPR95 (41.29 ± 1.2) and a higher AUROC (87.25 ± 0.9)—compelling

Table 5: OOD detection performance on large-scale unseen OOD data (CIFAR-100 InD, 300K Random Images as wild OOD, SVHN as test OOD), compared against wild-data baselines. Even under this stringent distributional-gap setting, Medix roughly halves the FPR of the best competing baseline while simultaneously achieving the highest AUROC.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

¹Since ResNet-34 is the standard architecture across these works, we adopt the same model to ensure consistency and facilitate a strictly fair comparison.

²<https://github.com>

evidence of its effectiveness at separating InD from OOD data even when the test-time OOD distribution is entirely unseen during training.

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

A further question concerns the role of pseudo-label quality $\hat{y}_{\tilde{x}_i}$: does the softmax confidence of the pseudo-labels materially affect OOD filtering, and could a simple pre-filtering step that discards low-confidence pseudo-labels improve robustness? To settle this question directly, we evaluate the effect of removing low-confidence pseudo-labels, using CIFAR-10 as the InD dataset and LSUN-Resize as the OOD dataset. We compute the softmax probability of each pseudo-labeled sample and discard predictions falling below a confidence threshold of 0.6; this removes 15.98% of the samples from the training set.

Performance is essentially unchanged by this intervention: both variants yield an identical FPR95 of 0.01%, and the filtering step nudges AUROC only marginally, from 99.98% to 99.99%. Removing low-confidence pseudo-labels therefore has no meaningful effect on either robustness or OOD detection, which conclusively demonstrates that Medix is resilient to noisy or low-confidence pseudo-labels and that additional pre-filtering is unlikely to yield appreciable gains.

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next address whether the Medix filtering phase is computationally feasible and memory-efficient at scale. To this end we profiled the procedure on an NVIDIA A100-SXM4-80GB GPU with roughly 15,000 unlabeled samples per InD–OOD pair, using CIFAR-10 and CIFAR-100 as the InD datasets and LSUN-Resize as the OOD dataset. As Table 6 reports, GPU memory usage remains modest and the filtering phase completes within a tractable wall-clock budget even for thousands of unlabeled samples. Medix filtering is thus computationally practical and imposes no excessive memory overhead, confirming its suitability for deployment on large datasets.

Table 6: Profiling of the Medix filtering phase on an NVIDIA A100 80GB GPU ($\approx 15,000$ unlabeled samples per pair): wall-clock time together with peak and current GPU memory for two InD–OOD pairs. Memory overhead remains under 100 MB throughout, confirming the practical scalability of the filtering procedure.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

Medix departs from recent semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) along three principal axes: objective, treatment of outliers, and technique. First, the objectives differ fundamentally. In OpenMatch Saito et al. (2021), the goal is to cope with outliers while training a standard classifier under semi-supervised learning (SSL); in our setting, by contrast, the goal is to detect OOD samples within unlabeled wild data and to train a dedicated OOD detector. Whereas SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) seek accurate classification in spite of outliers, Medix targets OOD detection itself in an open-world regime entirely devoid of labeled OOD data.

Second, the two paradigms treat outliers in diametrically opposite ways. SSL methods train a classifier robust to OOD samples by treating outliers as noise to be suppressed, thereby improving SSL performance; Medix instead explicitly detects OOD samples and actively exploits them as a training signal to construct a more effective and reliable OOD detection system.

Third, the underlying techniques diverge. OpenMatch rests on consistency regularization—enforcing agreement across stochastic transformations of the same input, formalized as soft constraints on the

771 classifier’s outputs that encourage smooth decision boundaries. Medix instead performs median-based
772 filtering, exploiting the statistical robustness of the median and its insensitivity to extreme values to
773 identify candidate OOD samples; its accompanying analysis derives error bounds governed by the
774 contamination effect (the OOD proportion) and the concentration effect (the sub-Gaussian behavior
775 of InD gradients), thereby guaranteeing that OOD detection error remains provably low.

776 **B Broader Impacts and Limitations**

777 As machine learning systems proliferate, robust OOD detection has become indispensable to reliable
778 model performance in real-world applications. Our median-centric framework directly advances
779 this goal, strengthening OOD handling and, with it, overall model safety. The significance of this
780 contribution extends beyond its theoretical guarantees to practical deployment across high-stakes
781 domains such as healthcare, autonomous systems, and finance: by improving OOD detection, Medix
782 addresses a central obstacle to trustworthy deployment and thereby fosters broader confidence in, and
783 adoption of, machine learning technologies. Promising avenues for future work include coupling
784 Medix with generative models for outlier synthesis and adapting it to dynamic environments in which
785 the OOD distribution drifts over time.

786 One limitation of Medix is its reliance on a mixture of unlabeled InD and OOD data that may, in
787 practice, be noisy, corrupted, or inconsistently sampled. This assumption is by no means peculiar to
788 our method; it reflects the intrinsic uncertainty of real-world deployment and constitutes a standard
789 starting point in robust learning theory. Critically, we render our assumptions explicit and verifi-
790 able—formalizing them in the main theorems and stating a precise geometric separation condition in
791 Remark 4.3—and we relax them substantially in Theorem C.3, which dispenses with sub-Gaussian
792 tails and establishes the resilience of our framework under considerably weaker distributional as-
793 sumptions. Taken together, these results demonstrate that the simplifying assumptions of Medix are
794 at once interpretable and relaxable, constituting a principled foundation for future advances in OOD
795 detection under realistic noise.

796 C Proof of Theorem

797 For clarity and consistency across the theoretical and empirical sections, Table 7 consolidates the key
 798 notation employed throughout the paper, covering gradient-related quantities, sample sizes, filtering
 799 terms, and the parameters that enter our concentration bounds. Unless stated otherwise, we denote
 800 vectors in boldface, write $\mathcal{X}$ for the input space, and compute all gradients with respect to a fixed
 801 pretrained model f_ϕ .

Table 7: Notation summary. We use bold symbols for vectors and $\mathcal{X}$ to denote the input space.

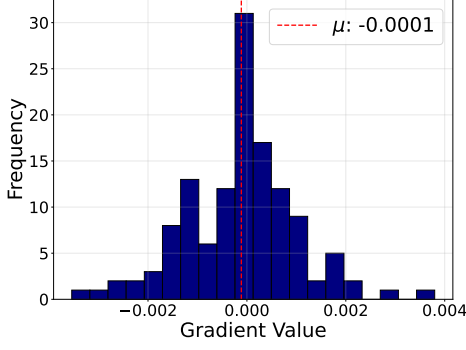
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

802 C.1 Bound on the Inlier Misclassification Rate

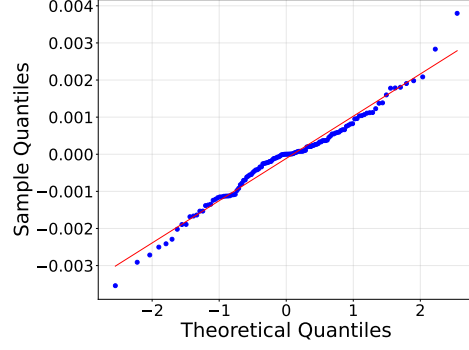
803 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 804 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 805 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 806 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

807 *Proof.* The argument proceeds in two steps. In **Step 1** we establish a high-probability concentration
 808 bound for the gradients of InD samples about their mean $\bar{\nabla}_{\text{in}}$, designating an InD sample whose
 809 gradient falls within this high-probability region a *good inlier*. In **Step 2** we invoke a “swapping”
 810 argument on the EWM objective to establish that excluding more than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of
 811 good inliers—where η denotes the fraction of InD samples with gradients outside the high-probability
 812 region—would contradict the optimality of the minimizer. Combining the two steps yields the stated
 813 upper bound on the inlier misclassification rate ERR_{in} .



(a) InD gradient histogram.



(b) InD gradient Q-Q plot.

Figure 4: Empirical evidence that InD sample gradients are coordinate-wise sub-Gaussian. (Left) The histogram of gradient values (CIFAR-100 InD data) concentrates tightly around the mean with light tails, exactly as expected under sub-Gaussianity. (Right) The Q-Q plot of empirical InD gradient quantiles against theoretical Gaussian quantiles aligns closely with the reference line, confirming sub-Gaussian behavior. Jointly, these two panels directly justify the sub-Gaussian assumption underlying Theorems 4.1 and 4.2.

Step 1: Concentration of InD Gradients. Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the gradient (with respect to ϕ) of a loss evaluated at an InD sample $\mathbf{x}$. Suppose each coordinate of this gradient is sub-Gaussian with variance proxy σ^2 , centered at the same mean vector $\bar{\nabla}_{\text{in}}$. That is, for each coordinate $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

It is straightforward to show that

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

so if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$ then there must exist at least one coordinate j with $|v_j| > \epsilon$.

Hence, combining (6) across $j = 1, \dots, d$ via a union bound yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the total number of InD points in the set $\mathcal{S}_{\text{wild}}$. We define an InD point $\mathbf{x}$ to be *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

Our goal is to show that, with high probability, only a small fraction of these m_{in} inliers can be bad.

From our sub-Gaussian assumption and applying the union bound across the d coordinates, we define a constant $p \in [0, 1]$ as follows:

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

Here, $\mathbf{x}$ is a generic InD sample, and we assume that the samples are i.i.d. across the m_{in} inliers.

Let X be the random variable denoting the number of bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define the indicator variable

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

829 Then, X can be expressed as

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

830 By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

831 Observing that X is a sum of m_{in} Bernoulli(p) random variables, each bounded between 0 and 1, we
832 can apply Hoeffding's inequality to obtain, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

833 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

834 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

835 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

836 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

837 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
838 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
839 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
840 called *good inliers*.

841 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 842 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

843 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 844 thereby contradicting the optimality of $\mathcal{S}^*$.

845 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 846 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

847 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

848 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

849 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (26)$$

850 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 851 available for swapping in.

852 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 853 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

854 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

855 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

856 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

857 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

858 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 859 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

860 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

861 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

862 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

863 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

864 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

865 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
866 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

867 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
868 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
869 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

870 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

871 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
872 increase it, so the median distance from μ_j does not increase and possibly decreases.

873 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

874 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

875 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
876 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

877 □

878 C.2 Bound on the Outlier Retention Rate

879 We now establish a non-asymptotic upper bound on the fraction of OOD points retained in the subset
880 returned by the MEDIX filter. The result parallels the inlier misclassification bound (Appendix C.1)
881 but, rather than relying on inlier concentration, exploits a geometric separation assumption between
882 the OOD and InD gradient means.

883 **Theorem C.2** (Outlier Retention Bound under Vector Separation). Assume that for every OOD point
 884 $x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:

885 **(I) Sub-Gaussianity.** Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .

886 **(II) Vector Mean Separation.** The mean OOD gradient vector μ_{out} satisfies

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d}, \quad \text{for some } \Delta > 0.$$

887 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

888 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

889 *Proof. Step 1: Concentration of OOD Gradients.*

890 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
 891 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

892 has probability at most $p_{\text{out}}(\varepsilon)$.

893 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

894 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

895 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

896 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

897 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

898 which holds with probability at least $1 - \delta$.

899 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

900 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_{\varepsilon}^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1-\pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

Thus, more than half the points in $\mathcal{S}_{\text{in}}^*$ are separated OODs whose gradients lie outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}^*}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}^*}) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}^*})\|_2^2. \quad (49)$$

This contradicts the optimality of $\mathcal{S}_{\text{in}}^*$, and so the assumption must be false. Therefore, our assumption must be false. With probability at least $1 - \delta$, we conclude that the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1-\pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^2 = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x)))_j - \bar{\nabla}_{\text{in},j}]^4 \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

921 **Differences with the original bound.** The original, sub-Gaussian bound carries a decay term

$$2 \exp\left(-\frac{\epsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

922 that vanishes exponentially in the dimension d , rendering the bound exceptionally tight for large d .
 923 The present bound, by contrast, invokes only finite fourth moments and thus exhibits the milder decay
 924 term

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2},$$

925 which decays as $O(1/d)$ and is correspondingly looser at large d . The trade-off is therefore fully
 926 explicit: the original bound requires sub-Gaussian tails, whereas the present one tolerates substantially
 927 heavier tails under a mere finite-fourth-moment condition. Crucially, both bounds retain the identical
 928 finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

929 which shrinks as m_{in} grows irrespective of the tail assumption—so the core robustness guarantee
 930 persists intact across both regimes.

931 *Proof.* The proof mirrors the structure of the original Theorem 4.1 and again comprises two steps:
 932 (1) bounding the fraction of “bad” InD samples (those far from ∇_{in}) via concentration, now without
 933 invoking sub-Gaussianity, and (2) bounding the inlier misclassification rate through a swapping
 934 argument. The sole substantive modification is that the sub-Gaussian tail bound is replaced by a
 935 moment-based bound derived from the finite-fourth-moment assumption, which is precisely what
 936 enables the generalization to heavier-tailed gradient distributions.

937 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x as:

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

938 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 939 and we assume ∇_{in} is the population mean. An InD sample is classified as “bad” if:

$$\|v\|_2 > \epsilon\sqrt{d}, \quad (52)$$

940 and “good” otherwise. Our goal is to bound the fraction of bad inliers among the m_{in} InD samples in
 941 S_{wild} .

942 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we analyze the ℓ_2 norm via the sum of squared coordinates. Assume that
 943 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$, each with:

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

944 Thus:

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

945 Compute the variance of v_j^2 :

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

946 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

947 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

948 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

949 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

950 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

951 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
952 bad with probability at most p . Then:

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

953 Since $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ if the i -th InD sample is bad and 0 otherwise, and I_i are i.i.d.
954 Bernoulli(p) variables bounded in $[0, 1]$, apply Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

955 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

956 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

957 The fraction of bad inliers is:

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

958 This η represents the upper bound on the fraction of InD samples whose gradients deviate from $\bar{\nabla}_{\text{in}}$
 959 by more than $\epsilon\sqrt{d}$, relying only on finite fourth moments and coordinate independence.

960 **Step 2: Contradiction via a Swapping Argument** We demonstrate that if $\mathcal{S}^*$ excludes an excessive
 961 number of InD, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

962 then it is possible to construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 963 thereby contradicting the optimality of $\mathcal{S}^*$.

964 Let m_ϵ be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 965 lies outside the closed ℓ_2 ball

$$B_\epsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \epsilon\sqrt{d} \right\}. \quad (69)$$

966 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

967 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

968 Define the swap size as

$$m'_\epsilon := \min \{m_\epsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

969 This quantity counts how many good inliers (those inside B_ϵ) are currently excluded from $\mathcal{S}^*$ and
 970 available for swapping in.

971 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ϵ points from $\mathcal{S}^*$ that lie outside B_ϵ (OOD points or bad inliers)
 972 and inserting m'_ϵ good inliers from B_ϵ in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

973 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\epsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

974 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\epsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

975 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

976 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

977 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 978 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

979 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

980 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta) m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

981 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

982 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

983 so the points lying inside B_ε form a strict majority in $\hat{\mathcal{S}}$.

984 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
 985 gradients in $\mathcal{S}^*$ and $\hat{X}^{(j)} = \{\hat{x}_1^{(j)}, \dots, \hat{x}_m^{(j)}\}$ the same in $\hat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

986 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
 987 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
 988 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\hat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

989 which implies

$$|\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\hat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

990 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
 991 increase it, so the median distance from μ_j does not increase and possibly decreases.

992 Thus, component-wise:

$$|\text{EWM}_j(G_{\hat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

993 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

994 This contradicts the assumption that $\mathcal{S}^*$ is a minimizer of the optimization problem, which concludes
 995 the proof.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

996 □